
NEPPLURAL: A BENCHMARK DATASET AND FRAMEWORK FOR EVALUATING PLURALISTIC ALIGNMENT IN NEPALI LARGE LANGUAGE MODELS

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ABSTRACT

As Large Language Models (LLMs) are deployed across low-resource linguistic regions, standard alignment techniques often suffer from "perspective collapse", forcing systems to mirror a singular dominant cultural or socio-political viewpoint. This study introduces NepPlural¹, a socio-technical framework and, to our knowledge, the first benchmark dataset for modelling and evaluating pluralistic alignment in the Nepali language. We focus on the contested socio-economic discourse surrounding Nepali youth migration and brain drain, a domain that encompasses competing values such as economic necessity, familial obligation, systemic frustration, and patriotism. We curate a corpus of code-mixed (Devanagari, Romanized Nepali, and English) public discourse drawn from Nepali YouTube channels, and we develop a four-axis annotation taxonomy that captures a commenter's Intent (stance on migration), Primary Driver, Value Orientation, and Affect. Labels are produced by three LLM annotators and merged by majority vote, with three-way ties resolved by a deterministic rule-based heuristic; we then verify a 20% sample with an LLM-as-a-Judge and manual inspection, which flags 35.2% of rows as containing at least one not-fully-defensible label, a figure that, on inspection, is largely driven by off-topic comments and quantifies the difficulty of the task. Fine-tuning five multilingual and Nepali-specific encoder baselines to predict the four axes yields a best mean macro-F1 of 0.52 (XLM-RoBERTa), with per-axis analysis revealing that Value Orientation is the most learnable axis while Intent, despite high accuracy, attains the lowest macro-F1 as models collapse to its majority class. NepPlural provides an evaluation resource and baseline suite to support the development of LLMs that better represent the socio-cultural realities of the Global South.

Keywords Large Language Models (LLMs), Pluralistic Alignment, Steerable Pluralism, Low-Resource NLP, Nepali, Data Annotation, Human-in-the-Loop, LLM-as-a-Judge

1 Introduction

As Large Language Models (LLMs) achieve widespread adoption globally, ensuring these systems serve users with diverse cultural values and perspectives has emerged as a critical open research problem. Standard post-training alignment techniques, such as Reinforcement Learning from Human Feedback (RLHF) and Direct Preference Optimization (DPO), optimize models to output a single, universally acceptable "ground truth." While effective for fact-based queries, this monolithic approach to alignment can lead to value collapse when models interact with non-Western demographics, resulting in culturally insensitive responses and an inability to accurately represent localized viewpoints [16].

¹<https://github.com/jinunyachhyon/NepPlural>

To address this, recent literature has proposed the framework of Pluralistic Alignment, arguing that AI systems should faithfully represent the diverse, conflicting values of a pluralistic society [16]. A crucial mechanism for achieving this is Steerable Pluralism; the ability of a model to use in-context information to override its internal priors and reliably steer toward specific, localized perspectives [16] [17]. However, recent findings indicate that standard instruction tuning can degrade a model’s in-context steerability, causing it to collapse into a narrow mode of generic, Western-centric responses [17]. While large-scale resources like the Spectrum Tuning [17] have been introduced to evaluate steerability, empirical benchmarks testing these principles in underrepresented languages and highly localized socio-political contexts of the Global South remain scarce.

To address this gap, we introduce NepPlural, which, to our knowledge, is the first benchmark and dataset designed to evaluate Steerable Pluralism within the Nepali cultural context. We focus our dataset on the socio-economic issue of Nepali youth migration and brain drain. This topic provides a suitable testbed for steerability because it encompasses a wide range of competing values and perspectives, including systemic frustration, economic constraints, familial obligation, and patriotism.

We construct NepPlural using a corpus of code-mixed Nepali digital discourse scraped from public YouTube videos. Moving beyond binary sentiment analysis, we develop a multi-dimensional annotation taxonomy to categorize the perspectives of Nepali citizens across four distinct axes:

- **Intent:** Stance on migration.
- **Primary Driver:** The main factor motivating the expressed viewpoint.
- **Value Orientation:** The underlying values reflected in the opinion.
- **Affect:** The emotional tone conveyed in the text.

Our primary contributions are as follows:

- **The NepPlural Dataset:** Pluralistic-alignment dataset for the Nepali language, comprising 2,148 code-mixed (Devanagari, Romanized Nepali, and English) public comments on youth migration and brain drain, labelled by three LLM annotators with majority voting and rule-based tie resolution. Each comment is annotated across four orthogonal persona axes.
- **A Multi-Dimensional Taxonomy:** A four-axis framework (*Intent*, *Primary Driver*, *Value Orientation*, and *Affect*) for capturing the competing socio-economic perspectives underlying a comment rather than its surface sentiment.
- **A Reproducible Annotation-and-Verification Pipeline:** A three-model majority-voting procedure with rule-based conflict resolution, followed by LLM-as-a-Judge verification and human inspection. We demonstrate that this verification step also functions as a quality filter by surfacing off-topic comments that do not express a migration-related persona.
- **Encoder Baselines and Error Analysis:** Fine-tuned NLU baselines for all four annotation axes, establishing a best mean macro-F1 score of 0.52. Our results show that multilingual encoders outperform current Nepali-specific models on code-mixed input, highlighting a measurable performance gap on real-world Nepali digital discourse.

By releasing this dataset, pipeline, and baseline suite, NepPlural provides an evaluation resource for systems that aim to represent the socio-cultural realities of the Global South, and lays the groundwork for directly benchmarking the in-context steerability of generative LLMs on Nepali personas.

2 Literature and Related Work

The alignment of large language models (LLMs) with human values has emerged as a central challenge in artificial intelligence research. As these systems expand from basic text generation to high-stakes decision-making domains, ensuring they reflect diverse perspectives becomes critical. This review synthesizes recent academic literature across several interconnected areas: the limitations of conventional alignment methods, steerability frameworks, evaluating pluralism, cultural value representation, and the use of LLMs for annotation and evaluation.

2.1 The Limitations of Conventional Alignment Methods

2.1.1 RLHF’s Scalar Reward Problem

Reinforcement Learning from Human Feedback (RLHF) has dominated LLM alignment since its introduction [10]. The approach shapes model behavior through scalar rewards reflecting general human preferences, typically emphasizing helpfulness and harmlessness [2]. However, this scalar reward structure creates a fundamental constraint: it can only reflect user preferences on average, failing to capture the complexity of individual human values which are often inconsistent, ambiguous, or conflicting [1].

A critical sociotechnical critique exposes deeper contradictions in RLHF’s alignment goals. Lindstrom et al. [11] demonstrate that the standard triad of honesty, harmlessness, and helpfulness contains inherent tensions. User-friendliness may encourage deception, flexibility can reduce interpretability, and safety measures sometimes limit useful flexibility. These tensions suggest RLHF cannot simultaneously optimize all three goals without trade-offs that remain unexamined in current implementations. The authors argue RLHF neglects ethically-relevant issues including the trade-offs between competing values, calling for auditing and disclosure standards to improve societal oversight.

2.1.2 DPO’s Fundamental Limitations

Direct Preference Optimization (DPO) emerged as a reward-free alternative that directly optimizes policies through classification objectives on human comparison data [15]. Despite achieving state-of-the-art results on academic benchmarks, theoretical analysis reveals DPO may have fundamental limitations. Xu et al. [23] conducted both theoretical and empirical studies showing DPO’s algorithmic properties create constraints that PPO does not share. The disconnect between DPO’s benchmark performance and its theoretical limitations suggests the method may not generalize as well as assumed.

Cal-DPO (Calibrated Direct Preference Optimization) addresses one specific limitation through reward calibration. Xiao et al. [22] demonstrate that calibrating implicit rewards to ensure comparability with ground-truth rewards in scale substantially improves alignment. This finding suggests DPO’s original formulation may suffer from reward scale mismatches rather than fundamental architectural flaws. Still, the need for calibration highlights that DPO is not as straightforward or theoretically sound as its simplified formulation implies.

2.1.3 The Average Preference Collapse

Standard alignment procedures exhibit a systematic tendency toward preference collapse. Sorensen et al. [16] argue that conventional alignment reduces distributional pluralism in models. When models are trained on aggregated preference data, they converge toward averaged responses that fail to represent the full spectrum of reasonable human perspectives. This phenomenon creates "Overton window narrowing," where models become less capable of representing diverse viewpoints within culturally acceptable bounds.

2.2 Methods for Steerability

2.2.1 Defining Steerable Pluralism

Steerability refers to the capacity of large language models (LLMs) to adopt specific perspectives, moral frameworks, or value systems in response to user instructions. Padolsey [13] conceptualizes steerable pluralism as the ability of an LLM to shift its interpretive vantage point in accordance with explicit prompts, thereby distinguishing it from static notions of alignment. Building on this idea, Adams et al. [1] formalize Steerable Pluralistic Models (SPMs) as systems designed to reliably condition their outputs on specified attribute profiles while maintaining internal consistency and responsiveness to contextual cues.

This concept extends beyond basic prompt engineering into broader frameworks of representational diversity and value conditioning. Sorensen et al. [16] further delineate three forms of pluralism in language models: Overton pluralistic models, which present a spectrum of socially or epistemically acceptable responses; steerable pluralistic models, which dynamically reflect specified perspectives upon request; and distributionally pluralistic models, which are calibrated to mirror population-level distributions of beliefs or values.

2.2.2 Few-Shot Comparative Regression and Modular Pluralism

Recent work has proposed a range of methods for implementing steerability in more efficient and flexible ways. Adams et al.’s [1] SPM uses few-shot comparative regression, computing the Euclidean distance between predicted attribute scores and target attribute vectors rather than requiring the LLM to directly generate or select responses. This

distance-based optimization reduces certain forms of model bias while enabling rapid re-alignment to different value specifications.

In a complementary direction, Feng et al. [6] propose Modular Pluralism, a framework that integrates multiple smaller community-based language models into a larger base system. During inference, the base model interacts with these community models to improve responsiveness to specified demographic values, yielding performance gains of up to 26.6% in value alignment tasks. Although computationally demanding, this modular approach enables continual adaptation as societal norms and values evolve over time.

Additionally, inference-time alignment strategies such as Personalized Alignment at Decoding-time (PAD) [3] and UserAlign [14] provide dynamic mechanisms for tailoring model outputs to individual users without the extensive data requirements associated with traditional reinforcement learning from human feedback (RLHF).

2.2.3 Cultural Alignment Evaluation

Evaluating cultural pluralism requires moving beyond Western-centric benchmarks. Zahraei et al. [24] introduce MENA Values as an evaluation framework for assessing alignment with Middle Eastern and North African cultural values, identifying phenomena such as cross-lingual value shifts and logit leakage. Similarly, Kharchenko et al. [9] demonstrate that while large language models can distinguish Hofstede’s cultural dimensions across 36 countries, they often fail to consistently apply these values in practical advice scenarios.

Empirical evidence further highlights demographic biases in model steerability. Models exhibit 9.7% lower steerability when prompted with incongruent personas compared to congruent ones, frequently reverting to stereotypical responses [12].

2.2.4 Existing Gap in Nepali Language

While recent research has focused on the technical challenges of Nepali language processing, such as the morphological complexity and data scarcity identified by Subedi et al. [18], the field has yet to address the sociotechnical challenges of LLM alignment. Specifically, the ‘English-centric reasoning’ observed in low-resource contexts [18] suggests that current alignment methods (RLHF/DPO) may not merely underperform in Nepali but may inadvertently impose Western moral frameworks on the language. Furthermore, the failure of LLMs to utilize structural linguistic knowledge, as highlighted by recent experiments with grammar-based prompting, indicates that we lack the ‘steerable pluralism’ frameworks necessary to preserve the unique socio-cultural and honorific nuances of the Nepali language.

As argued by Gabriel et al. [7], preference modeling tends to converge toward homogenized moral reasoning patterns, especially when supervision signals are aggregated across culturally diverse inputs. This raises concerns that “helpfulness” and “harmlessness” criteria may reflect culturally situated judgments rather than neutral ethical principles, particularly in low-resource and culturally distinct languages such as Nepali. In this sense, alignment is increasingly recognized not only as a technical optimization problem but also as a form of cultural encoding.

Parallel developments in interpretability and controllability research suggest that current instruction-tuning approaches lack mechanisms for preserving linguistic and sociocultural structure. Work on controllable generation has shown that while prompting and reinforcement learning can adjust surface-level style and tone, they fail to consistently preserve deeper syntactic and pragmatic constraints, such as honorific systems, evidentiality, and formality shifts [25]. This limitation is particularly relevant for Nepali, where honorific agreement and politeness marking are structurally embedded in grammar and discourse rather than being purely lexical choices.

Taken together, these strands of research suggest a clear gap at the intersection of alignment, multilinguality, and sociolinguistic fidelity. While technical improvements continue to enhance multilingual benchmark performance, there remains limited understanding of how alignment processes reshape cultural and linguistic expression in low-resource languages. Addressing this gap requires moving beyond performance-centric evaluation toward frameworks that explicitly model cultural-linguistic integrity as a first-class objective in LLM alignment.

Crucially, existing Nepali language resources target lower-level tasks such as named-entity recognition and part-of-speech tagging [18]; none provide annotated data for the competing values, personas, or steerable perspectives that pluralistic alignment requires. To our knowledge, NepPlural is the first dataset to address this gap for Nepali.

2.3 LLM-based Annotation and Evaluation

A growing body of work uses large language models as annotators and as evaluators in place of, or alongside, human labellers. Gilardi et al. [8] show that instruction-tuned LLMs can match or exceed crowd workers on several text-annotation tasks at a fraction of the cost, while single-model labels can inherit that model’s biases and vary across

prompts. Aggregating multiple independent annotators by majority vote is a standard remedy for such variance, and applying it across distinct LLM families reduces the chance that any single model’s idiosyncrasies dominate the final label. Aggregation strategies such as self-consistency [21] similarly show that combining multiple model outputs improves reliability over any single pass. In parallel, the LLM-as-a-Judge paradigm [26] uses a strong model to assess the quality of generated or labelled content, achieving high agreement with human judgments on many tasks while remaining far cheaper to scale. These methods are attractive for low-resource languages such as Nepali, where expert annotators are scarce, but they carry a known risk: the same Western-centric and English-centric priors documented in Section 2.2 can surface in both annotation and judging. We therefore treat LLM annotation and judging as a scalable first pass to be combined with majority voting, deterministic rule-based resolution, and human inspection (Section 3), rather than as a standalone source of ground truth.

3 Methodology

The execution architecture of the NepPlural framework follows a multi-stage pipeline designed to ensure data diversity, annotation quality, and replicability.

3.1 Data Collection Strategies and Scope

To build a corpus capturing multi-perspective public discourse on migration, we extracted text from the comment sections of high-engagement Nepali YouTube channels. We targeted three sources covering migration podcasts, interviews, and public commentary: IDS, Thaha Research, and The Nepali Comment, to span a wide range of grassroots viewpoints. This yielded informal, unstructured ideological stances expressed directly by digital participants, rather than curated or edited text.

3.2 Preprocessing and Anonymization

Raw Nepali digital text is fragmented by overlapping Unicode variants, mixed Devanagari/Romanized script, and irregular spacing, and as scraped public data it may contain personal information. The collected data therefore underwent a multi-stage preprocessing pipeline before annotation:

- **PII Anonymization:** because the raw scrape could contain personally identifiable information, we scrubbed personal details: names, contact information, and other personal identifiers, from the text, so that no comment is processed or released in a form that identifies an individual.
- **Unicode Normalization:** all text was processed with a regex-based normalization routine to enforce consistent Devanagari character representation and collapse overlapping Unicode variants.
- **Segmentation:** sentence boundaries were identified and tokenized using the IndicNLP Library, an open-source toolkit designed for Indo-Aryan languages.
- **Content Filtering:** we then applied a deliberately conservative keep/drop filter. Rather than normalize linguistic variation away, which would erase exactly the code-mixed character we want to study, a comment is dropped only if it carries no classifiable content; everything else is retained verbatim. The filter removes empty, emoji-only, or punctuation-only comments (no word tokens); very short comments below a two-token minimum (e.g. "wow", "nice", "ok"); comments consisting only of URLs, @-mentions, or hashtags; spam (channel promotion, "subscribe", "sub4sub", promo links); and exact duplicate texts (first occurrence kept). HTML entities and tags are unescaped and stripped before these checks. Each kept comment is enriched with metadata for later analysis: script (Devanagari / Roman-Nepali / code-mixed / Latin), language, emoji count, word count, and source, so results can be sliced by script type.

We deliberately do not apply lexical or script normalization beyond the steps above, so Devanagari, Romanized Nepali, code-mixed, and English comments survive in their original form. After preprocessing and filtering, the final corpus comprises 2,148 annotated comments.

3.3 Annotation Schema Design

To map the multi-perspective discourse around Nepali youth migration, we move away from binary sentiment toward a four-axis taxonomy. Every instance is annotated across four orthogonal axes:

- **Intent (stance on migration):** Pro-Migration, Anti-Migration, Trapped/Regretful, or Neutral/Observation.

- **Primary Driver (root cause):** Economic Necessity, Family Obligation, Systemic/Political Anger, or Patriotism/Love.
- **Value Orientation (who/what is prioritized):** Collectivist-Family, Collectivist-Nation, or Individualist-Self.
- **Affect (emotional tone):** Despairing/Sad, Angry/Frustrated, Hopeful/Motivated, or Pragmatic.

3.4 Multi-Annotator Labelling, Majority Voting, and Rule-Based Resolution

Each comment was independently labelled across all four axes by three LLM annotators: GPT-5.2, Gemini 3.5 Flash, and Sonnet 4.6, using a shared instruction prompt. For each axis we combine the three labels by majority vote: where at least two annotators agree on an axis, that label is adopted. When all three annotators disagree on an axis (a three-way tie), the cell is flagged and resolved by a deterministic, domain-specific keyword heuristic over the comment text rather than discarded, producing the Final Annotations. The heuristic uses migration-domain keyword lists per axis: for Intent, departure/return phrases decide Pro- vs Anti-Migration, defaulting to Neutral/Observation; for Primary Driver, political/government terms map to Systemic/Political Anger, family terms to Family Obligation, job/money terms to Economic Necessity, and love-of-country terms to Patriotism/Love, with Economic Necessity as the dataset-dominant fallback; Value Orientation and Affect are resolved analogously, falling back to Individualist-Self and Pragmatic respectively. Meta comments (about the video itself), language-choice remarks, and factual corrections are overridden to Neutral/Observation, since they express no migration stance. We note this resolution step is automated, not human; human inspection enters at the verification stage (Section 3.5).

3.5 LLM-as-a-Judge and Manual Verification

To estimate label reliability, we drew a 20% sample (429 comments) and re-checked it with an independent LLM-as-a-Judge, combined with manual human inspection of the flagged, low-confidence rows. A row is marked "wrong" if any one of its four labels is judged not defensible against the comment text. This conservative, all-axes-must-hold criterion provides an upper bound on the per-row error rate. On manual inspection of the flagged rows we found that most are off-topic or generic comments (e.g. "well done", "nice video") that express no migration persona. We deliberately did not remove these comments from the dataset — the verification was a confidence check, not a filtering pass — but the finding explains the headline flag rate (Section 4.1).

3.6 NLU Baseline Fine-Tuning

To establish how learnable the taxonomy is, we fine-tune five transformer encoders as multi-task classifiers that jointly predict the four axes: multilingual BERT (bert-base-multilingual-cased), XLM-RoBERTa (base), two Nepali-specific models from IRIIS-RESEARCH (BERT 110M and RoBERTa 125M), and NepBERTa. All models share the same configuration: seed 42, 10 epochs, learning rate 1e-5, batch size 16, and maximum sequence length 192 tokens. The corpus is split 1503/215/430 for train/validation/test, and the checkpoint with the best validation mean macro-F1 is used for testing. We report per-axis macro-F1 and accuracy plus the mean macro-F1 across axes.

3.7 Off-Topic and No-Persona Comments

Public comment sections contain many reactions directed at the video rather than the migration topic — generic praise, channel shout-outs, emoji-only replies, for which the four NepPlural axes do not meaningfully apply. We deliberately retain these comments in the released dataset rather than removing them: the 20% verification (Section 3.5) was a confidence check, not a filtering pass, and the LLM-as-a-Judge reliably surfaces such comments because no persona label can be justified from their text (quantified in Section 4.1). Keeping them, with their best-effort labels and verification metadata, lets future users either model this category explicitly or filter it themselves; a dedicated "No-Persona / Off-Topic" label is a natural extension we leave to future work.

4 Results and Discussion

We evaluate NepPlural along two axes: the reliability of the annotation pipeline (Section 4.1) and the learnability of the four-axis taxonomy by fine-tuned encoder baselines (Section 4.2). We do not yet benchmark the in-context steerability of generative LLMs; that evaluation is set up by the released prompts and left to future work (Section 5).

4.1 Label Reliability under LLM-as-a-Judge Verification

Across the 429-comment verification sample, the judge flagged 151 rows (35.2%) as containing at least one not-fully-defensible label. Because a row fails if any one of its four axes is judged wrong, this figure is a strict upper bound on per-row error. Crucially, manual inspection of the flagged rows shows that most failures are not genuine annotation errors but a property of the source data: 68% of the flagged rows (103/151) are short or non-substantive comments that carry no migration persona at all, generic praise and off-topic remarks such as "Excellent explanation", "KEEP MAKING VIDEOS", and "♡♡♡♡ weldone brother" (mean length of a flagged comment is 9.3 words). For such comments, any assignment along the four persona axes is ill-defined, so the judge correctly marks the labels as not defensible. The 35.2% figure therefore conflates two distinct phenomena: true ambiguity on substantive comments, and the unavoidable noise of social-media comment sections, rather than indicating a 35% labelling-error rate.

Table 1: LLM-as-a-Judge verification (Opus 4.8) on a 20% sampled subset of the dataset.

Source / File	Judged	Wrong	Wrong Rate (%)
IDS - migration old podcast	59	11	18.6
IDS - regular video comments	211	74	35.1
Thaha Research	40	13	32.5
The Nepali Comment - Illegal Migr.	78	39	50.0
The Nepali Comment - Migration	41	14	34.2
Total	429	151	35.2

The wrong rate varies sharply by source, from 18.6% on the long-form migration podcast to 50.0% on the "Illegal Migration" comment set. This spread tracks the density of non-substantive comments: sources dominated by short reactions and generic praise yield higher flag rates, while the long-form podcast, whose comment section is more reflective, yields the lowest. The variation thus reflects how on-topic a source’s audience is, and supports treating off-topic comment filtering as part of the pipeline rather than reading the rate as a model error.

4.2 NLU Baseline Performance

Table 2 reports test-set performance for the five fine-tuned encoders. The multilingual models clearly outperform the Nepali-specific encoders: XLM-RoBERTa achieves the highest mean macro-F1 (0.516), followed by multilingual BERT (0.489), while NepBERTa (0.389), IRIIS-RESEARCH BERT (0.359), and IRIIS-RESEARCH RoBERTa (0.356) trail. The advantage of the multilingual models likely stems from their exposure to the code-mixed Devanagari/Romanized/English text that the corpus, which monolingual Nepali pretraining does not cover.

Table 2: Test mean macro-F1 ($n_{\text{test}} = 430$). Best per column in bold.

Model	Mean Macro-F1
XLM-RoBERTa (base) [4]	0.516
mBERT (cased) [5]	0.489
NepBERTa [20]	0.389
IRIIS-RESEARCH BERT 110M [19]	0.359
IRIIS-RESEARCH RoBERTa 125M [19]	0.356

Table 3: Per-axis macro-F1 comparison between top two models on the test set.

Axis	XLM-RoBERTa	mBERT
Intent	0.355	0.427
Primary Driver	0.512	0.479
Value Orientation	0.631	0.619
Affect	0.566	0.432

A revealing pattern emerges between accuracy and macro-F1 on the Intent axis: all models score high accuracy on Intent (0.72-0.78) but low macro-F1 (0.34-0.43). This gap indicates that the classifiers exploit a dominant Intent class

rather than separating the minority stances (e.g. Trapped/Regretful), so headline accuracy overstates true competence. Macro-F1 is therefore the appropriate metric for this imbalanced, pluralistic setting. Across all five models, Value Orientation is consistently the most learnable axis (macro-F1 up to 0.63). Intent, by contrast, is the weakest axis by macro-F1 for the two strongest models (0.35-0.43) despite its high accuracy, a direct consequence of the majority-class collapse noted above, while Affect and Primary Driver fall in between. No axis exceeds 0.64, underscoring the headroom the benchmark exposes.

4.3 Discussion

Two findings stand out. First, the judge’s 35.2% flag rate is largely explainable rather than alarming: 68% of flagged rows are off-topic or generic comments that do not express a migration persona, so the framework is correctly identifying the limits of where the taxonomy applies. This is a useful signal: it argues for an explicit "no persona / off-topic" handling step and shows that genuine disagreement on substantive comments is far lower than the headline rate suggests. Second, the best fine-tuned baseline reaches only ~0.52 mean macro-F1, leaving wide headroom that quantifies the gap NepPlural is designed to expose; the gap is real even after discounting the off-topic noise above. The relative weakness of Nepali-specific encoders against multilingual models on code-mixed input also suggests that monolingual Nepali pretraining alone is inadequate for real Nepali digital discourse, where Devanagari, Romanized Nepali, and English are freely mixed.

4.4 Limitations

Our study has several limitations. First, the corpus is single-domain, covering only youth migration and brain drain, and is drawn entirely from the comment sections of three Nepali YouTube channels; findings may not transfer to other contested socio-political domains or to other platforms. Second, tie-resolution in the annotation pipeline relies on deterministic keyword heuristics with a dataset-dominant fallback, which can introduce a systematic bias toward majority labels on the hardest, most ambiguous comments. Third, our reliability estimate uses a single LLM-as-a-Judge, which may carry its own cultural and linguistic biases; we mitigate but do not eliminate this through manual inspection of the sample. Finally, the empirical evaluation reported here uses fine-tuned encoder classifiers, not the generative, in-context steerability setup that the NepPlural framework ultimately targets; benchmarking frontier generative LLMs on the released personas remains future work.

5 Conclusion

This paper introduced NepPlural, a socio-technical framework and, to our knowledge, the first benchmark dataset for evaluating pluralistic alignment in the Nepali language. Focusing on the contested discourse of youth migration and brain drain, we curated a corpus of 2,148 code-mixed public comments and annotated each along four orthogonal axes: Intent, Primary Driver, Value Orientation, and Affect, that capture the persona and motivation behind a comment rather than its conformity to external moderation norms. Labels were produced by three LLM annotators and merged by majority vote, with three-way ties resolved by a deterministic rule-based heuristic and a 20% sample independently verified by an LLM-as-a-Judge and manual inspection.

Our evaluation yields two clear conclusions. First, the LLM-as-a-Judge step is diagnostic rather than damning: although it flagged 35.2% of the verification sample, 68% of those flags are off-topic or generic comments ("nice video", "inspirational", and the like) that carry no migration persona, so the headline rate mostly measures the noise inherent in social-media comment sections, not labelling error. This argues for treating off-topic filtering as a first-class pipeline stage. Second, the task remains genuinely hard for current models: five fine-tuned encoder baselines reached a best mean macro-F1 of only 0.52 (XLM-RoBERTa), and no single axis is reliably learned - Intent in particular attains high accuracy but the lowest macro-F1, as models collapse to its majority class. Multilingual encoders clearly outperformed Nepali-specific models on this code-mixed data, indicating that monolingual pretraining alone is not yet adequate for real Nepali digital discourse.

NepPlural contributes the dataset, a reproducible annotation-and-verification pipeline, and a baseline suite against which future systems can be measured. Future work includes (i) extending the corpus beyond migration to other contested Nepali socio-political domains; (ii) evaluating the in-context steerability of frontier generative LLMs prompted to adopt each NepPlural persona, using the provided evaluation prompt; and (iii) testing alignment methods that preserve the honorific and pragmatic structure of Nepali. By quantifying the gap between current models and faithful pluralistic representation, NepPlural aims to support the development of AI systems that better reflect the socio-cultural realities of the Global South.

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